

A disproof of Conjecture 00000000752

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Counterexample

Take $p = 3$ and the full one-sided shift on two symbols. A presentation by a one-vertex graph has adjacency matrix $A = [2]$. Its characteristic polynomial is $X - 2$.

Theorem 1. *The shift has entropy $\log 2$, whereas the conjectural p -adic-root formula gives 0.*

Proof. Every position of a word can independently be either symbol, so the number of admissible length- n words is 2^n . Consequently

$$h_{\text{top}} = \lim_{n \rightarrow \infty} \frac{1}{n} \log(2^n) = \log 2 > 0.$$

The only characteristic root is 2. It is a 3-adic unit, hence $|2|_3 = 1$. The formula in the conjecture therefore returns $\log |2|_3 = \log 1 = 0$. These values differ. $\square$

The issue is structural: real topological entropy is governed by the ordinary spectral radius, not by taking a p -adic norm of the characteristic roots.

Lean 4 certificate

The project in `lean4/` defines the full-shift word count as the finite product k^n , computes 2^n , certifies that 2 is a 3-adic unit of norm 1, and proves the predicted equality impossible at $n = 1$. No unproved placeholder or additional axiom is used.